

CHELTENHAM GIRLS HIGH SCHOOL



YEAR 11 YEARLY EXAMINATION 2009

Mathematics

General Instructions

- Working time: $1\frac{1}{2}$ hours.
- Write using black or blue pen.
- Board-approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Each **question** is to be started on a **new page**.
- Write your **name** on every page.

Total Marks - 75

- Attempt all Questions 1-5
- All questions are of equal value.

Name : _____ Class Teacher _____

Student Number : _____

Q1	Q2	Q3	Q4	Q5	Total	%
/15	/15	/15	/15	/15	/75	

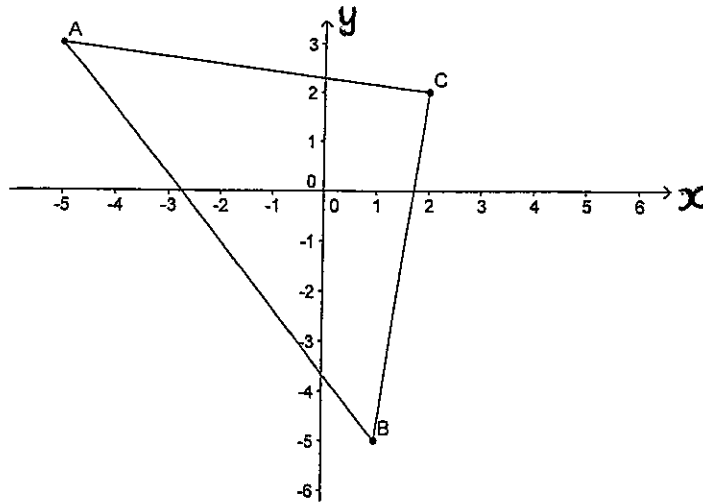
Answer each question on a new page

Question 1 (15 marks) Start on a new page	Marks
(a) Find, correct to two decimal places, the value of $\frac{(4.32)^2}{7.53 - 2.48}$	1
(b) Factorise $2x^2 + 7x - 15$	2
(c) Find integers a and b such that $(2 - \sqrt{3})^2 = a - b\sqrt{3}$	3
(d) Express $\frac{3x-2}{5} - \frac{x+1}{2}$ as a single fraction in its simplest form	2
(e) Given that $\cos \theta = \frac{-3}{4}$ and $\sin \theta > 0$, find the <u>exact</u> value of $\tan \theta$:	2
(f) Differentiate	
(i) $y = 6 - 3x^4$	1
(ii) $y = \frac{x+1}{x-2}$	2
(iii) $y = \sqrt{(3x-2)^3}$	2

Question 2 (15 marks) Start on a new page

Marks

- (a) A, B and C are three points on a number plane whose coordinates are $(-5,3)$, $(1,-5)$ and $(2,2)$ respectively.



- | | |
|---|---|
| (i) Calculate the length of AB | 2 |
| (ii) Find the equation of the line AB | 2 |
| (iii) The line through C, perpendicular to AB, meets AB at N.
Find the equation of CN | 2 |
| | |
| (b) Solve $2x^2 < x$ | 3 |
| | |
| (c) The equation $2x^2 - x + 4 = 0$ has roots α and β .
Find the values of: | |
| (i) $\alpha + \beta$ | 1 |
| (ii) $\alpha\beta$ | 1 |
| (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ | 2 |
| (iv) $\alpha^2 + \beta^2$ | 2 |

Question 3 (15 marks) Start on a new page

Marks

(a) Differentiate $y = (2x^2 + 3)(5 - x)$ **2**

(b) Solve $|2x - 4| = 3x - 1$ **3**

(c) The roots of the quadratic equation $mx^2 - 20x + n = 0$ are 3 and -5.
Find the values of m and n , where m and n are both integers. **3**

(d) A function is defined by the following rule

$$f(x) = \begin{cases} 0 & \text{if } x \leq -2 \\ -1 & \text{if } -2 < x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$

Find:

(i) $f(-3) + f(-1) - f(0)$ **2**

(ii) $f(a^2)$ **1**

(e) Sketch the region bounded by $x^2 + y^2 < 16$ and $2x - 3y + 6 \geq 0$ **4**

Question 4 (15 marks) Start on a new page

Marks

- (a) Find the centre and the radius of the circle with equation

3

$$x^2 + y^2 - 18y - 40 = 0$$

- (b) Solve $4^x - 5 \times 2^x - 24 = 0$

3

- (c) Find $\lim_{x \rightarrow 4} \frac{3x^2 - 48}{x - 4}$

2

- (d) Given $f(x) = 5x^2 - 15x - 12$

(i) Find the equation of its axis of symmetry.

1

(ii) Find the minimum value of the function

1

- (e) Find the exact value of :

(i) $\tan 60^\circ$

1

(ii) $\sin 240^\circ$

1

- (f) Find the values of k for which $f(x) = 3x^2 - (k+2)x + (k+2)$ is positive definite .

3

Question 5 (15 marks) Start on a new page

Marks

(a) The focus of a parabola is $(2, -5)$ and its directrix is the line $y = -1$

(i) Sketch the parabola and indicate the coordinates of the vertex V. 3

(ii) Write down the focal length of the parabola. 1

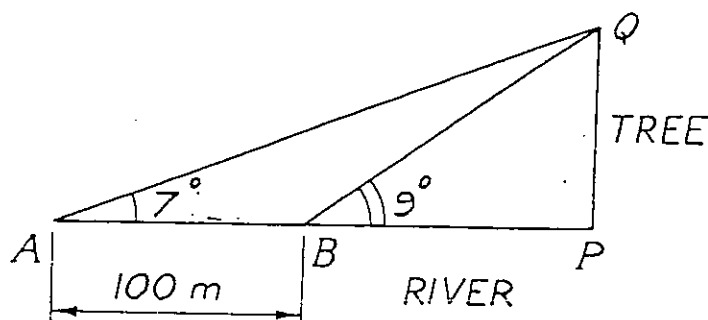
(iii) Find the equation of the parabola. 2

(b) The function $f(x)$ is defined by $f(x) = \frac{1}{x^2 + 1}$

(i) Find $f'(x)$ 2

(ii) Find the coordinates of the stationary point and determine its nature. 2

(c)



The diagram above was sketched by a surveyor, who measured the angle of elevation of a tree top on the other side of the river to be 7° at the point A . At point B , 100 metres directly towards the tree from A , the angle of elevation was 9° .

Let the height of the tree PQ be h and the width of the river BP be r

(i) Write an expression for r in terms of h . 1

(ii) Derive an expression for h independent of r . 3

(iii) Calculate the height of the tree correct to three significant figures. 1

END OF EXAM

Year 11 , Yearly Examination
solutions - 2009
Mathematics

Question 1

$$(a) \frac{(4.32)^2}{7.53 - 2.48}$$

$$= \frac{18.6624}{5.05}$$

$$= 3.70$$

$$(b) 2x^2 + 7x - 15$$

$$(2x - 3)(x + 5)$$

$$(c) (2 - \sqrt{3})^2 = a - b\sqrt{3}$$

$$4 - 4\sqrt{3} + 3 = a - b\sqrt{3}$$

$$7 - 4\sqrt{3} = a - b\sqrt{3}$$

$$a = 7, \quad b = 4$$

$$(d) \frac{3x-2}{5} - \frac{x+1}{2}$$

$$\frac{(3x-2)}{5} - \frac{(x+1)}{2}$$

$$= \frac{2(3x-2) - 5(x+1)}{10}$$

$$= \frac{6x - 4 - 5x - 5}{10}$$

$$= \frac{x - 9}{10}$$

$$(e) \cos \theta = -\frac{3}{4}, \sin \theta > 0$$

$$\tan \theta = -\frac{\sqrt{7}}{3}$$

$$(f) (i) y = 6 - 3x^4$$

$$\frac{dy}{dx} = -12x^3$$

$$(ii) y = \frac{x+1}{x-2}$$

$$\frac{dy}{dx} = \frac{(x-2) - (x+1)}{(x-2)^2}$$

$$= \frac{x - 2 - x - 1}{(x-2)^2}$$

$$= \frac{-3}{(x-2)^2}$$

$$(iii) y = \sqrt{(3x-2)^3}$$

$$y = (3x-2)^{\frac{3}{2}}$$

$$\frac{dy}{dx} = \frac{3}{2} (3x-2)^{\frac{1}{2}} \times 3$$

$$= \frac{9}{2} (3x-2)^{\frac{1}{2}}$$

$$= \frac{9\sqrt{3x-2}}{2}$$

Question 2

(a) (i) $A = (-5, 3), B = (1, -5)$

$$\begin{aligned} AB &= \sqrt{(1 - (-5))^2 + (-5 - 3)^2} \\ &= \sqrt{(6)^2 + (-8)^2} \\ &= \sqrt{100} \\ &= 10 \text{ units.} \end{aligned}$$

(ii) $m_{AB} = \frac{-5 - 3}{1 - (-5)} = -\frac{8}{6} = -\frac{4}{3}$

using point $A = (-5, 3)$

$$(y - 3) = -\frac{4}{3}(x - (-5))$$

$$3(y - 3) = -4(x + 5)$$

$$3y - 9 = -4x - 20$$

$$4x + 3y + 11 = 0$$

∴ Equation of line AB is $4x + 3y + 11 = 0$.

(iii) $m_{CN} = \frac{3}{4} \quad (CN \perp AB)$

using point $C = (2, 2)$

$$(y - 2) = \frac{3}{4}(x - 2)$$

$$4(y - 2) = 3(x - 2)$$

$$4y - 8 = 3x - 6$$

$$3x - 4y + 2 = 0$$

∴ Equation of line CN is $3x - 4y + 2 = 0$

(b) $2x^2 < x$

$$2x^2 - x < 0$$

$$x(2x - 1) < 0$$



$$0 < x < \frac{1}{2}$$

(c) $2x^2 - x + 4 = 0$

(i) $\alpha + \beta = -\frac{(-1)}{2} = \frac{1}{2}$

(ii) $\alpha\beta = \frac{4}{2} = 2$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$

$$= \frac{\beta + \alpha}{\alpha\beta}$$

$$= \frac{\frac{1}{2}}{2} = \frac{1}{4}$$

$$= \frac{1}{4}$$

(iv) $\alpha^2 + \beta^2$

$$= (\alpha + \beta)^2 - 2\alpha\beta$$

$$= \left(\frac{1}{2}\right)^2 - 2(2)$$

$$= \frac{1}{4} - 4$$

$$= -3\frac{3}{4}$$

Question 3

(a) $y = (2x^2 + 3)(5 - x)$

$$\begin{aligned}\frac{dy}{dx} &= 4x(5 - x) + (-1)(2x^2 + 3) \\ &= 20x - 4x^2 - 2x^2 - 3 \\ &= -6x^2 + 20x - 3\end{aligned}$$

(b) $|2x - 4| = 3x - 1$

$(2x - 4) = (3x - 1)$ or $(2x - 4) = -(3x - 1)$

$2x - 4 = 3x - 1$ $2x - 4 = -3x + 1$

$-x = 3$

$x = -3$

$5x = 5$
 $x = 1$

Check

$x = -3$

$|2x - 4| = 3x - 1$

$|2(-3) - 4| = 3(-3) - 1$

$|-10| = -10$

$10 = -10$

$\therefore x = -3$ is not a solution.

check

$x = 1$

$|2x - 4| = 3x - 1$

$|2(1) - 4| = 3(1) - 1$

$|2 - 4| = 3 - 1$

$1 - 2 = 2$

$2 = 3$ $x = 1$ is a solution.

$\therefore$ solution is $x = 1$

(c) $mx^2 - 20x + n = 0$

roots = α, β .

$\alpha + \beta = -\frac{(-20)}{m} = 3 + -5$

$\frac{20}{m} = -2$

$-2m = 20$

$m = -10$

$\alpha\beta = \frac{n}{m} = 3 \times (-5)$

$\frac{n}{m} = -15$

$\frac{n}{-10} = -15$

$n = +150$

$\therefore m = -10$ and $n = +150$.

(d) (i) $f(-3) + f(-1) - f(0)$

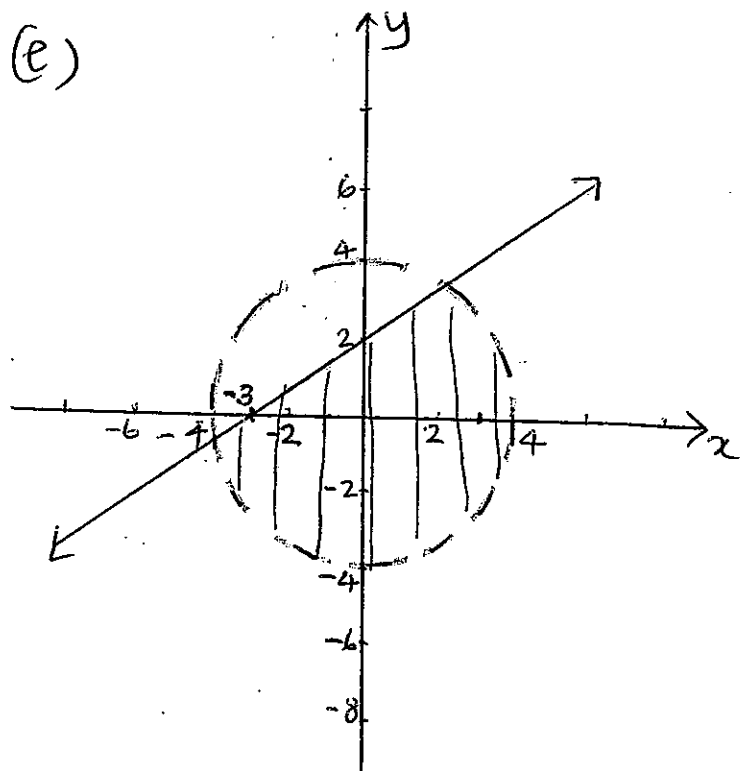
$= 0 + -1 + 0$

$= -1$

(ii) $f(a^2) = a^2$

since $a^2 > 0$

(e)



question 4

(a) $x^2 + y^2 - 18y - 40 = 0$

$$x^2 + y^2 - 18y + 81 = 40 + 81$$

$$(x-0)^2 + (y-9)^2 = 121$$

radius = 11 units.

centre = (0, 9)

(b) $4^x - 5 \times 2^x - 24 = 0$

$$(2^x)^2 - 5(2^x) - 24 = 0$$

take $y = 2^x$

$$y^2 - 5y - 24 = 0$$

$$(y+3)(y-8) = 0$$

$$y = -3 \text{ or } y = 8$$

$$2^x = -3 \text{ or } 2^x = 8$$

$$2^x \neq -3 \quad 2^x = 2^3$$

$\therefore x = 3$ is the solution.

(c) $\lim_{x \rightarrow 4} \frac{3x^2 - 48}{x - 4}$

$$= \lim_{x \rightarrow 4} \frac{3(x^2 - 16)}{x - 4}$$

$$= \lim_{x \rightarrow 4} 3(x+4)$$

$$= 3(4+4)$$

$$= 3 \times 8$$

$$= 24$$

(d) $f(x) = 5x^2 - 15x - 12$

(i) axis of symmetry:

$$x = -\frac{b}{2a}$$

$$x = -\frac{(-15)}{2(5)}$$

$$= \frac{15}{10}$$

$$x = \frac{3}{2}$$

(ii) minimum value

$$= 5\left(\frac{3}{2}\right)^2 - 15\left(\frac{3}{2}\right) - 12$$

$$= -23\frac{1}{4}$$

(e) (i) $\tan 60^\circ = \sqrt{3}$

(ii) $\sin 240$

$$= \sin(180 + 60)$$

$$= -\sin 60$$

$$= -\frac{\sqrt{3}}{2}$$

(f) $f(x) = 3x^2 - (k+2)x + (k+2)$

$$a = 3 > 0, \Delta < 0$$

$$\Delta = [-(k+2)]^2 - 4(3)(k+2) < 0$$

$$= k^2 + 4k + 4 - 12k - 24 < 0$$

$$= k^2 - 8k - 20 < 0$$

$$k^2 - 8k - 20 < 0$$

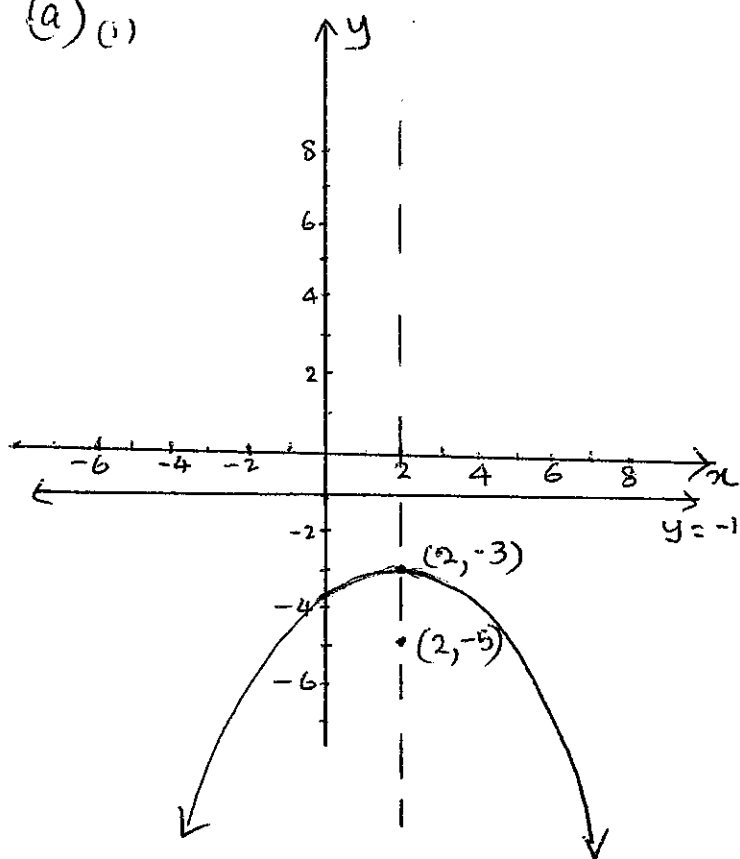
$$(k-10)(k+2) < 0$$



$$-2 < k < 10$$

Question 5

(a) (i)



vertex = $(2, -3)$

(ii) Focal length = 2
 $a = 2$

$$\begin{aligned} \text{(iii)} \quad (x-h)^2 &= -4a(y-k) \\ (x-2)^2 &= -4(2)(y+3) \\ (x-2)^2 &= -8(y+3) \end{aligned}$$

$$\begin{aligned} \text{(b) (i)} \quad f(x) &= \frac{1}{(x^2+1)} \\ f(x) &= (x^2+1)^{-1} \\ f'(x) &= -(x^2+1)^{-2} \times 2x \\ &= \frac{-2x}{(x^2+1)^2} \end{aligned}$$

(b)(ii) At stationary points

$$\begin{aligned} f'(x) &= 0 \\ \frac{-2x}{(x^2+1)^2} &= 0 \\ -2x &= 0 \\ x &= 0 \end{aligned}$$

x	-1	0	1
$f'(x)$	$-\frac{2}{4}$	0	$-\frac{2}{4}$
	/	=	\

The turning point is a maximum at $(0, 1)$

$$\text{(c) (i)} \quad \tan \theta^\circ = \frac{h}{r}$$

$$\therefore r = \frac{h}{\tan \theta^\circ}$$

$$\text{(ii)} \quad \text{using } \triangle BPQ \\ r = \frac{h}{\tan \theta^\circ}$$

$$\text{using } \triangle APQ \\ \tan 7^\circ = \frac{h}{r+100}$$

$$\begin{aligned} r+100 &= \frac{h}{\tan 7^\circ} \\ r &= \frac{h}{\tan 7^\circ} - 100 \end{aligned}$$

(C)
(ii) continued

$$r = \frac{h}{\tan 9^\circ} \rightarrow (1)$$

$$r = \frac{h}{\tan 7^\circ} - 100 \rightarrow (2)$$

$$(2) = (1)$$

$$\frac{h}{\tan 7^\circ} - 100 = \frac{h}{\tan 9^\circ}$$

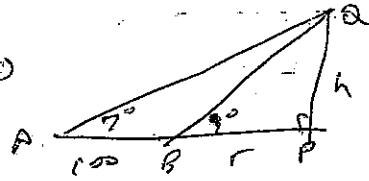
$$\frac{h}{\tan 7^\circ} - \frac{h}{\tan 9^\circ} = 100$$

$$[\tan 9^\circ - \tan 7^\circ]h = 100 \times \tan 9^\circ \times \tan 7^\circ$$

$$h = \frac{100 \tan 9^\circ \tan 7^\circ}{[\tan 9^\circ - \tan 7^\circ]}$$

(iii) 54.6 m

2) (i)



$$\begin{aligned} \tan 9^\circ &= \frac{h}{r} \\ \therefore r &= \frac{h}{\tan 9^\circ} \end{aligned}$$

$$\begin{aligned} \text{In } \triangle APQ, \tan 7^\circ &= \frac{h}{100 + r} \\ &= \frac{h}{100 + \frac{h}{\tan 9^\circ}} \end{aligned}$$

$$\tan 7^\circ \left(100 + \frac{h}{\tan 9^\circ} \right) = h$$

$$2h = 100 \tan 7^\circ + h \frac{\tan 7^\circ}{\tan 9^\circ}$$

$$h \left(1 - \frac{\tan 7^\circ}{\tan 9^\circ} \right) = 100 \tan 7^\circ$$

$$h = \frac{100 \tan 7^\circ}{1 - \frac{\tan 7^\circ}{\tan 9^\circ}}$$

$$= 54.627 \dots \approx 54.6 \text{ m}$$